

Lezione 2 – Studi di funzione

Matematica del continuo

Modulo 11 - Studio di funzioni

Unità didattica 3 – Funzioni trigonometriche

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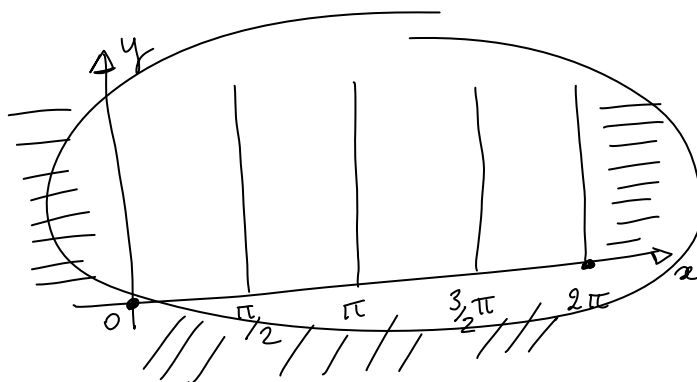
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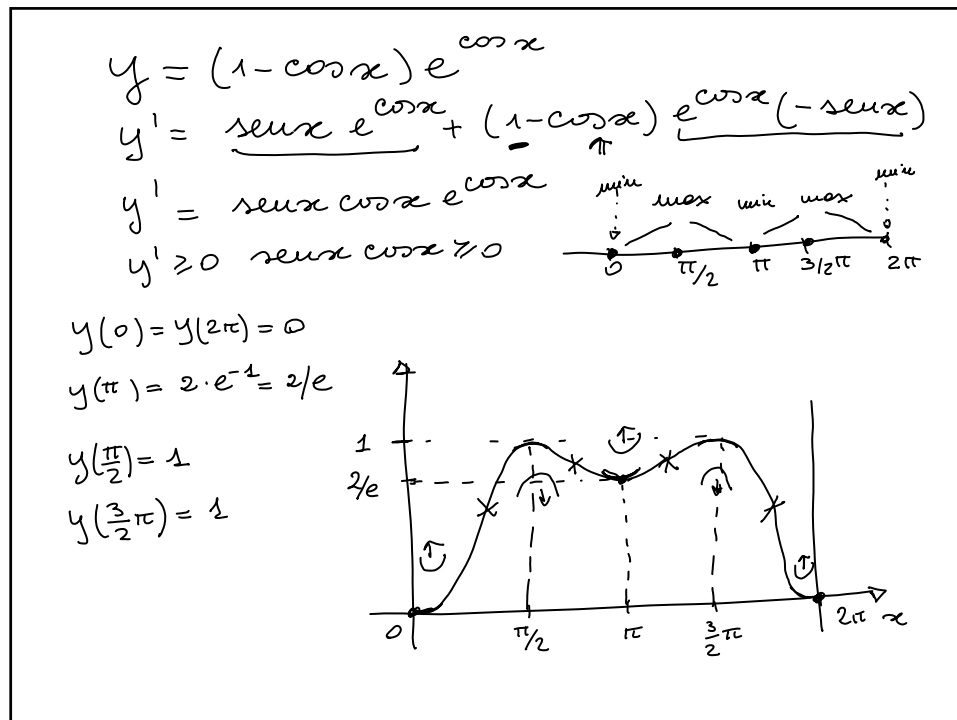
$$y = (1 - \cos x) e^{\cos x} \quad CE : \mathbb{R}$$

$$[0, 2\pi]$$

$$y(0) = y(2\pi) = 0$$

$$y \geq 0 \quad 1 - \cos x \geq 0 \quad \begin{cases} = 0 & \text{per } x = 0, 2\pi \\ > 0 & \text{altrimenti} \end{cases}$$





$$y' = \sin x \cdot \cos x \cdot e^{\cos x}$$

$$y'' = \cos x \cdot \cos x e^{\cos x} + \sin x (-\sin x) e^{\cos x} + \sin x \cos x e^{\cos x} \cdot (-\sin x) =$$

$$= e^{\cos x} \left(\underbrace{\cos^2 x - \sin^2 x}_{\cos 2x} - \underbrace{\sin x \cos x \sin x}_{\frac{1}{2} \sin 2x} \right)$$

$$= \cancel{\sin} e^{\cos x} \left(\cos 2x - \frac{1}{2} \sin 2x \cdot \sin x \right)$$

$$y'' = 0 \quad \frac{1}{2} \sin 2x \sin x = \cos 2x$$

$$\textcircled{\tan 2x = \frac{2}{\sin x}} \quad \textcircled{\neq 0}$$

